AI Customer Support Platform

SCRUM ARTIFACTS

How we will build the AI Customer Support Platform

Product Goal, backlog, Sprints, and what "done" means. Press → to start.

Space or arrows to move · Esc for overview · S for speaker notes · F for full screen · I for all decks

THE ONE SENTENCE EVERYTHING SERVES

Product Goal

A customer can contact an organization through **website chat**, receive **accurate knowledge-grounded answers**, complete at least one **authorized read-only business lookup**, and **reach a human without losing context**.

The architecture must be ready to add **voice** without duplicating business reasoning.

FIT

Why Scrum suits this product

AI behaviour is uncertain

We cannot predict answer quality on paper. Each Sprint ends with a demo and an evaluation run, so we inspect real behaviour.

Vendors drive risk

Voice and business integrations depend on provider limits and credentials. Short Sprints and spikes expose that early.

Security is not a phase

Tenant isolation, authorization and privacy are inside every Definition of Done, not a final checkpoint.

ACCOUNTABILITIES

Three accountabilities, one team

Product Owner

- Product value and Product Goal
- Orders the Product Backlog
- Accepts Sprint results

Scrum Master

- Coaches Scrum
- Removes impediments
- Helps the team improve

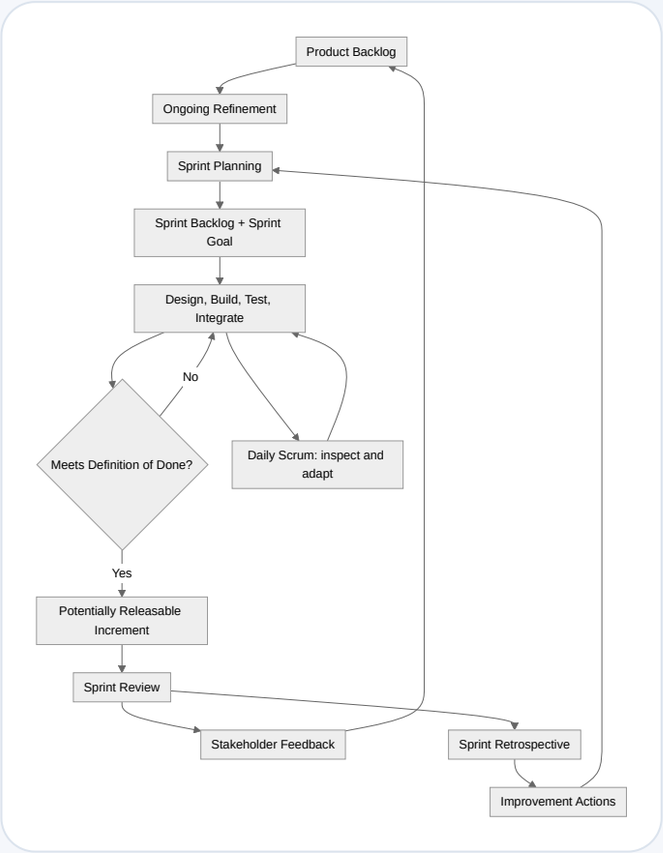
Developers

- Cross-functional team
- Design, build, test, integrate
- Secure and operate the Increment

Stakeholders: support teams, organization admins, customers, security and privacy, business-system owners. Roles may be held by a small team, but the accountabilities stay clear.

How work flows

- **Refine** the Product Backlog all the time
- **Plan** a Sprint: one Sprint Goal, selected items
- **Build** until each item meets the Definition of Done
- **Review** the Increment with stakeholders, feed learning back into the backlog
- **Retrospect** and carry improvement actions into the next Sprint



TWO-WEEK SPRINTS

The rhythm of a Sprint

1. **Sprint Planning**
   Pick a Sprint Goal and backlog items
2. **Daily Scrum**
   15 minutes, inspect and adapt
3. **Build to Done**
   Design, code, test, secure
4. **Sprint Review**
   Demo working software
5. **Retrospective**
   One or two real improvements

Refinement runs all the time: split, clarify, reorder and estimate upcoming work. It is not a formal event.

43 USER STORIES

Twelve epics

E01 Tenant and access
3 stories

E02 Conversation core
4 stories

E03 Website chat
3 stories

E04 Knowledge and RAG
4 stories

E05 AI orchestration
4 stories

E06 Safe business tools
4 stories

E07 Human handoff
4 stories

E08 Tickets
2 stories

E09 Voice MVP
5 stories · Release 2

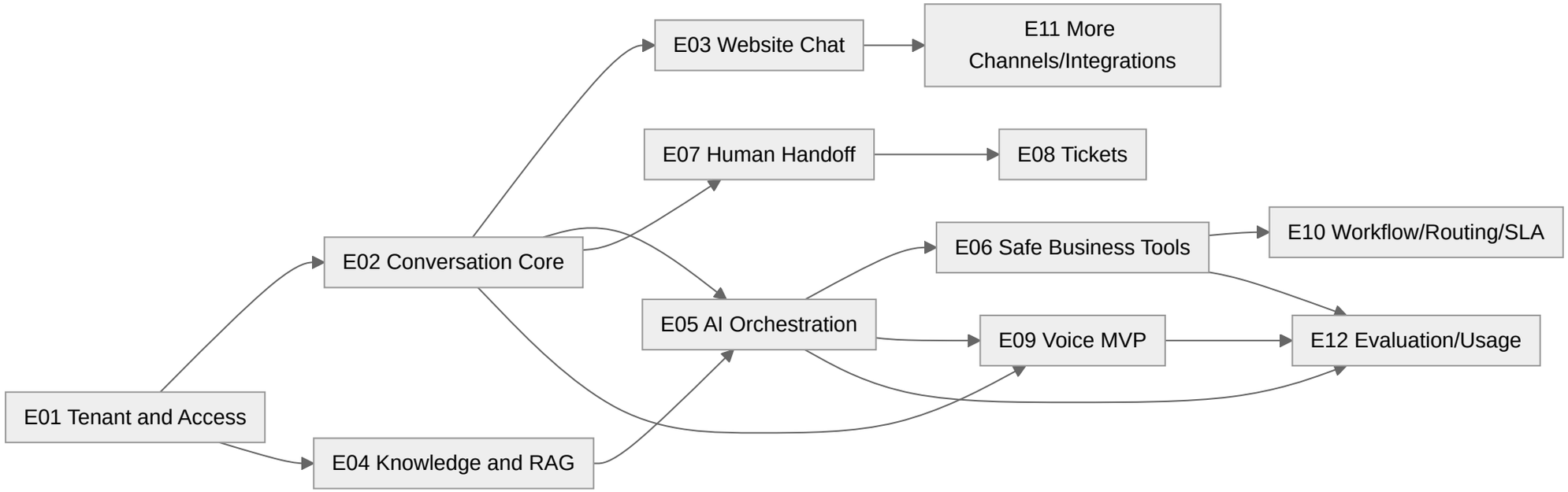
E10 Workflow, routing, SLA
3 stories · Release 3

E11 Integrations and channels
3 stories · Release 3

E12 Observability, evaluation, usage
4 stories

Blue = Release 1 (text MVP) · Amber = Release 2 (voice) · Green = Release 3 (operations)

What depends on what



Tenant and access come first. AI orchestration needs conversation core and knowledge.

SUGGESTED ORDER, REORDER WITH EVIDENCE

What we build first

1. Tenant and access foundation
2. Conversation core
3. Website chat
4. Minimum observability and security, in parallel
5. Knowledge ingestion and RAG
6. AI orchestration
7. One safe read-only business tool
8. Human handoff
9. Ticket basics
10. Voice
11. Workflows, routing, SLA
12. More connectors and channels, advanced evaluation

Order changes with user research, risk, technical dependencies and Sprint Review feedback.

THREE RELEASES

Where the Sprints lead

Release 1

Text support MVP

Tenant foundation, website chat, grounded answers, one read-only tool, human handoff, tickets, audit and telemetry.

Release 2

Voice MVP

Real-time browser voice on the same engine, speech processing, transcript sync, interruption and voice handoff.

Release 3

Operations expansion

More channels, workflow automation, routing and SLA, connectors, stronger evaluation, usage controls.

PART TWO

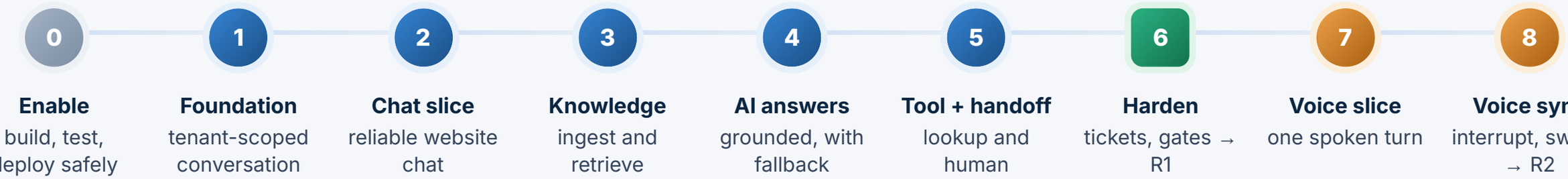
The Sprint plan

Nine Sprints, each ending with a usable Increment and a demo.

02

A PLANNING HYPOTHESIS, REFORECAST AFTER SPRINT 1

Nine Sprints at a glance



Sprint 0 only if needed, and never a permanent phase. Security, observability and tests ship inside every Sprint.

Sprints 1 and 2 · Secure foundation and website chat

Sprint 1

Goal: a tenant-scoped conversation and message can be created, persisted and retrieved.

- US-001 Organization foundation
- US-003 Tenant isolation tests
- US-010 Conversation creation
- US-011 Message persistence

Demo: create, append, fetch timeline, prove cross-tenant access is denied.

Sprint 2

Goal: a visitor exchanges text messages with the backend reliably.

- US-020 Widget configuration
- US-021 Chat session
- US-022 Reconnect and resume
- US-013 Inbound deduplication

Demo: send a message, see it persisted and returned, replay a duplicate, drop the network.

Sprints 3 and 4 · Knowledge and grounded answers

Sprint 3

Goal: organization knowledge is ingested and retrieved within tenant boundaries.

- US-030 Source registration and upload
- US-031 Async parse, chunk and index
- US-032 Retrieval with source metadata
- Ingestion status and retry

Demo: upload a document, watch status, query it, see source references.

Sprint 4

Goal: chat answers common questions from approved knowledge and handles missing evidence safely.

- US-040 Turn orchestration
- US-033 Grounded answer or fallback
- US-041 Follow-up context
- US-042 Model failure handling
- Minimal evaluation set and usage recording

Demo: grounded FAQ, follow-up, unknown fallback, provider timeout.

Sprints 5 and 6 · Tool, handoff, release hardening

Sprint 5

Goal: one authorized read-only lookup, and transfer to a human.

- US-050 Tool registry and allowlist
- US-051 Order lookup
- US-053 Tool-call audit
- US-060 Inbox queue
- US-061 Customer-requested handoff
- US-062 Handoff summary

Demo: tool success, denial and failure, takeover with context.

Sprint 6 → Release 1

Goal: unresolved work is tracked and the MVP meets release gates.

- US-070 Ticket creation
- US-071 Ticket filtering
- US-043 Agent publish and rollback
- US-110 End-to-end tracing
- US-112 First regression evaluation
- Security, load, backup and runbooks

Demo: conversation becomes a ticket; metrics, audit, test report, rollback.

Sprints 7 and 8 · Voice

Sprint 7

Goal: validate the voice architecture and complete one browser voice turn.

- Choose speech strategy and provider
- US-080 Start and end a session
- US-081 Speak and hear a reply
- Voice telemetry and fallback
- Security and consent review

Demo: speak, hear a reply, transcript linked to the conversation.

Sprint 8 → Release 2

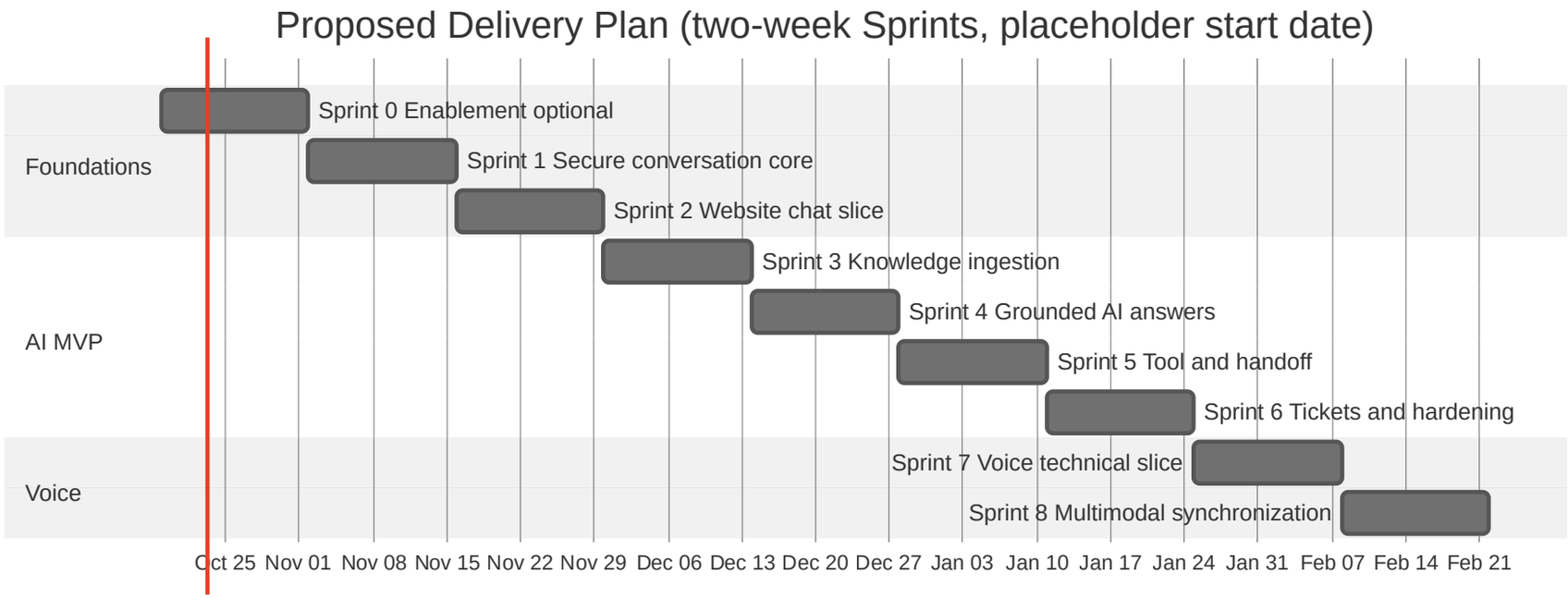
Goal: voice and text share one coherent timeline.

- US-082 Interruption
- US-083 Switch between text and voice
- US-084 Voice handoff for the chosen provider
- Voice quality regression scenarios
- Cost and latency dashboard

Demo: speak, interrupt, type a follow-up, transfer to human.

Timeline

placeholder dates



Replace the start date with the team's real Sprint 0 or Sprint 1 start. Capacity is unknown, so the sprint count may change.

PART THREE

Quality: ready, done, safe to ship

The agreements that keep the Increment trustworthy.

TEAM AGREEMENT, NOT A GATE FOR DISCOVERY WORK

Definition of Ready

- The user or business outcome is clear
- Acceptance criteria are testable
- Dependencies and provider assumptions are identified
- UX, API and data behaviour are understood, security and privacy considered
- Small enough for one Sprint, and estimated
- Credentials, test data and environments are available or planned

THREE LAYERS

Definition of Done

Every item

- Acceptance criteria met
- Reviewed, module boundaries kept
- Tests incl. failure cases
- Tenant isolation and authz tested
- Safe, redacted telemetry
- Docs and contracts updated
- Rollback or flag understood

AI items add

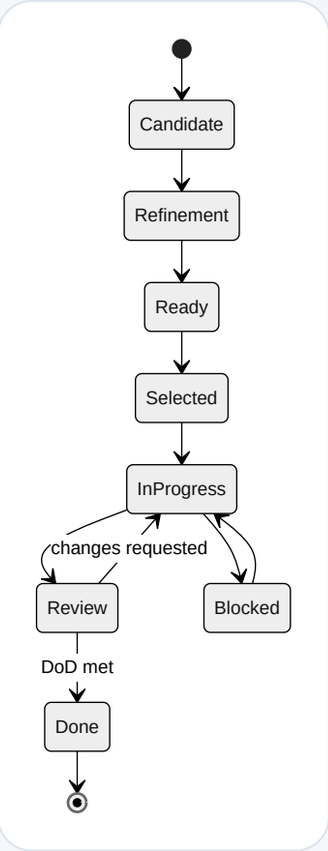
- Model and prompt version identifiable
- Tool permissions enforced in code
- Retrieval tenant scoped
- Empty and uncertain paths tested
- Regression example added
- No secrets or hidden reasoning in logs

Voice items add

- Mic permission and active state clear
- Cleanup on disconnect
- Interruption tested
- Transcript order tested
- Retention per policy
- Latency and failure metrics

A story's life

- **Candidate → Refinement → Ready:** clarified until it meets the Definition of Ready
- **Selected:** pulled into a Sprint at planning
- **In progress ↔ Review:** changes requested send it back
- **Blocked:** visible, and returns to In progress when freed
- **Done:** only when the Definition of Done is met



EXIT CRITERIA

When a release may ship

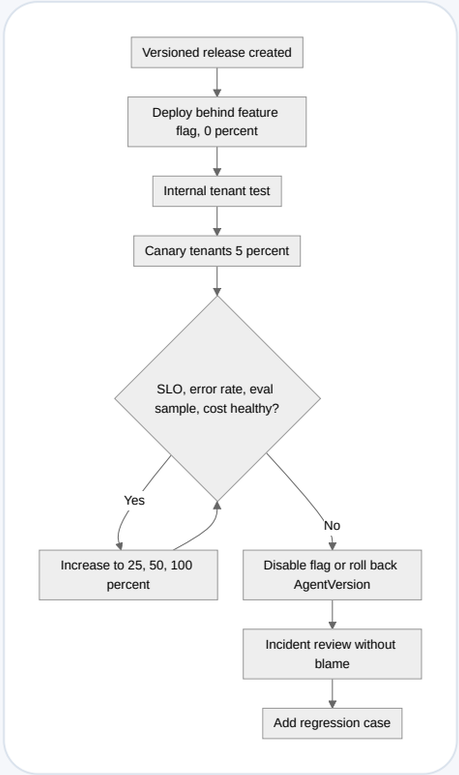
Release 1

- Tenant isolation tests pass
- Inbound messages deduplicated
- Safe fallback for unknown questions
- Tools allowlisted, authorized, audited
- A human can take over
- API and worker telemetry available
- No open critical evaluation failure

Release 2

- Session start, stop, cleanup reliable
- Transcript finalization and ordering
- Interruption behaviour tested
- Audio and transcript privacy implemented
- Latency and cost metrics visible
- Handoff validated for the chosen provider

Roll out safely, roll back fast



Feature flag, canary tenants, then a gradual increase. Unhealthy means disable the flag or roll back the agent version.

TOP RISKS AND WHAT WE DO ABOUT THEM

Risks and mitigations

RISK	MITIGATION
Prompt injection	Tool allowlist, server-side authorization, untrusted-content boundaries, adversarial tests
Tenant data leakage	Tenant-scoped indexes and filters, cross-tenant tests in CI
LLM latency and cost	Budgets, timeouts, model routing, measurement
Duplicate webhook or tool call	Durable inbox, idempotency keys, deduplication
Voice interruption races	Sequence ids, partial vs final, cancellation semantics
Unsupported AI answer	Evidence thresholds, safe fallback, handoff
Provider lock-in, scope too broad	Adapters and contract tests; text-MVP vertical slice first

DECIDE THE RIGHT THINGS AT THE RIGHT TIME

Decide now, defer on purpose

Architecture decisions to write (ADRs)

- Modular monolith boundaries
- Database and tenant isolation
- Queue, outbox and inbox
- Vector and search technology
- LLM abstraction and data policy
- Voice strategy, transport, retention
- Authentication and customer verification
- Tool authorization and approval

Do not decide yet

- One microservice per module
- A universal workflow builder
- Several speech providers at once
- Autonomous write access to every system
- Permanent raw audio retention
- Complex billing before usage is known

RESOLUTION QUALITY, NOT ONLY SPEED

How we know it works

Correct

Answer rate on a curated evaluation set

Grounded

Source support rate

Resolved

Task completion, escalation rate by reason

Fast and safe

p50 and p95 latency, tool success, availability

Plus customer satisfaction and cost per resolved conversation.

WHAT WE NEED TO START

Open items

- **People:** name the Product Owner and Scrum Master, confirm team size and capacity
- **Calendar:** pick the first Sprint start date and Sprint length (default 2 weeks)
- **Estimates:** size the 43 stories in team-specific points
- **Access:** LLM, speech and a commerce or order API with test credentials
- **Backlog gaps:** schedule US-002, US-012, US-052, US-063 (needed for Release 1 but not yet in a Sprint)

Questions?

Next decks: **product.html** for what we are building, **technologies.html** for the technologies, **engineering.html** for how.

Full detail: rapport.pdf